

VERONA VILLAFRANCA, Italy

WMO: 160900

Lat: 45.396N

Lon: 10.889E

Elev: 73

StdP: 100.45

Time Zone: 1.00 (EUC)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-4.9	-3.2	-14.1	1.1	5.1	-10.1	1.6	4.1	9.1	7.0	7.9	6.2	1.4	0	0.416

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	10.7	34.1	22.6	32.8	22.5	31.2	21.9	25.8	29.1	24.8	28.6	23.9	28.1	2.5	230

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
25.0	20.3	26.9	23.9	18.9	26.1	22.8	17.7	25.5	79.8	29.6	75.5	29.3	72.0	28.3	30.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 8.4	7.0	5.7	-7.2	35.6	2.1	1.7	-8.7	36.8	-10.0	37.7	-11.1	38.7	-12.7	39.9		
(5)			-7.7	27.4	2.1	1.5	-9.2	28.5	-10.4	29.3	-11.6	30.2	-13.1	31.3		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	14.1	3.2	5.1	9.4	13.4	18.3	22.8	24.9	24.5	19.8	14.4	8.8	3.9		
	DBStd	8.25	2.73	2.90	3.15	3.09	3.04	3.22	2.60	2.80	2.92	3.13	3.53	3.25		
	HDD10.0	660	210	138	48	7	0	0	0	0	0	4	62	191		
	HDD18.3	2186	468	370	278	149	37	4	0	1	18	126	286	448		
	CDD10.0	2153	0	1	28	109	259	383	463	448	293	141	26	1		
	CDD18.3	637	0	0	0	2	38	137	204	191	61	4	0	0		
	CDH23.3	6254	0	0	2	32	332	1337	2172	1921	442	15	0	0		
	CDH26.7	2414	0	0	0	2	76	498	922	813	102	1	0	0		
(14) Wind	WSAvg	2.1	1.7	2.1	2.4	2.5	2.4	2.3	2.1	2.0	2.0	1.8	1.9	1.6		
(15) Precipitation	PrecAvg	769	44	43	53	66	77	81	61	86	70	78	73	52		
	PrecMax	1096	175	168	149	178	203	181	143	389	185	237	189	140		
	PrecMin	293	0	0	1	12	0	6	0	0	0	0	0	2		
	PrecStd	160	37	37	37	31	45	44	36	70	49	58	39	34		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	12.7	16.2	22.1	26.1	31.2	34.5	36.0	36.2	31.1	25.1	18.6	13.7		
		MCWB	9.1	10.2	12.1	15.9	19.3	22.7	22.1	22.8	20.9	17.2	14.6	9.7		
	2%	DB	10.2	13.7	19.1	24.0	28.9	33.0	34.0	34.1	29.3	23.0	16.7	11.2		
		MCWB	7.1	8.6	11.9	14.9	18.6	22.3	22.8	22.5	20.1	17.6	13.9	8.6		
	5%	DB	9.0	11.9	17.2	22.1	26.9	31.2	32.7	32.8	27.8	21.2	15.2	10.0		
		MCWB	6.5	7.6	11.1	14.0	17.6	21.5	22.8	22.5	19.3	16.5	13.2	8.0		
	10%	DB	7.8	10.1	15.3	20.1	25.0	29.8	31.1	31.0	26.0	19.8	14.0	8.5		
		MCWB	5.9	7.0	10.1	13.5	17.1	20.9	22.1	22.1	18.5	15.8	12.3	7.0		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	10.6	11.6	14.5	17.6	22.6	26.1	27.2	27.3	23.7	20.7	17.0	11.9		
		MCDB	11.8	15.2	18.5	22.7	26.1	29.0	30.0	30.3	26.2	22.3	17.5	12.6		
	2%	WB	8.4	9.9	13.1	16.1	20.7	24.6	25.8	25.8	22.4	19.0	15.0	9.7		
		MCDB	9.2	11.9	17.0	21.0	25.0	28.6	29.3	28.9	25.7	20.8	15.8	10.4		
	5%	WB	7.1	8.8	12.1	15.2	19.5	23.6	24.8	24.8	21.3	17.9	13.8	8.3		
		MCDB	8.1	10.5	15.8	19.9	23.7	28.1	28.7	28.6	25.0	20.0	14.9	9.1		
	10%	WB	6.1	7.8	11.1	14.4	18.5	22.6	23.9	23.8	20.4	16.9	12.7	7.4		
		MCDB	6.9	9.3	14.2	18.6	22.7	27.3	28.3	27.9	23.9	18.8	13.9	8.1		
(35) Mean Daily Temperature Range	5% DB	MDBR	7.2	8.5	9.8	10.1	10.2	10.3	10.7	10.7	10.1	8.4	7.0	6.9		
		MCDBR	9.3	11.7	13.2	13.5	12.9	11.8	12.0	12.5	12.0	10.4	7.9	8.9		
	5% WB	MCWBR	6.5	7.2	7.1	5.9	5.1	5.1	5.2	5.5	5.1	5.7	5.6	6.4		
		MCWBR	7.3	9.3	11.2	11.4	11.3	11.0	10.8	11.2	10.3	8.4	6.1	6.4		
(40) Clear-Sky Solar Irradiance	taub		0.348	0.377	0.422	0.457	0.456	0.481	0.464	0.453	0.441	0.430	0.380	0.342		
	taud		2.377	2.271	2.150	2.086	2.138	2.116	2.170	2.216	2.218	2.202	2.307	2.405		
	Ebn at Noon		744	786	797	803	817	797	805	799	771	712	690	710		
	Edh at Noon		77	102	131	152	149	153	144	132	122	108	82	68		
(44) All-Sky Solar Radiation	RadAvg		1.42	2.23	3.53	4.50	5.57	6.37	6.53	5.58	4.12	2.50	1.48	1.22		
	RadStd		0.22	0.33	0.48	0.49	0.52	0.34	0.35	0.37	0.28	0.35	0.21	0.16		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		+0.53	N/A	N/A	+0.93	N/A	N/A	N/A	-111	+162	+102
(47) Regional (49 neighbors)		+0.34	N/A	N/A	+0.56	N/A	N/A	N/A	-93	+65	+10

Nomenclature: See separate page